Sai Krishna

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Work experience

CGI Jun 2026 - Jul 2026

Software Engineering Intern *Bengaluru, India*

- Built and deployed a Python automation platform on the Atlassian Jira REST API that automated timesheet collection and submission, reducing manual effort by 80%, saving 15+ analyst-hours per week, and eliminating manual reporting errors.
- Designed an SLA monitoring and escalation engine with configurable threshold rules for 500+ active tickets, reducing SLA breaches by 60% and improving on-time ticket resolution by 35%.
- Applied object-oriented design to create modular Python packages for Jira issue lifecycle management, including triage, assignment, transitions, and closure, enabling new workflows without code duplication.
- Analyzed throughput, breach frequency, and sprint-velocity data to identify operational bottlenecks and reduce process cycle time by 25%.

Projects

NEXUS | *Python, HNSW, SimHash, Vector Embeddings, Fast Cluster Routing*

- Built an in-process HNSW index and two-level SimHash and cluster-routed semantic cache for retrieval over 18,000+ documents, achieving 0.81ms lookups versus 1.51ms baseline (1.9× faster) with 0.11ms warm-cache p50.
- Implemented entropy-adaptive similarity thresholds that improved cache classification F1 score by +56% (0.889 vs. 0.571 baseline) while preserving 0.8330 NDCG@10 retrieval quality.

PowerBench | *PyTorch, Qwen3, QLoRA, GRPO, CUDA, Python*

- Benchmarked LLM inference and GRPO reasoning RL training (Qwen3-0.6B, QLoRA rank-16) on consumer GPU hardware (RTX 2050, 4GB VRAM), profiling latency, throughput, and power draw across FP16/INT8/INT4 precisions.
- Implemented TV-distance attention head scoring across 448 heads (28 layers × 16) to identify reasoning-sensitive heads for pruning and interpretability analysis.

Parameter Golf | *PyTorch, Quantization, Tied Embeddings, FineWeb*

- Trained a GPT-style language model for OpenAI's Parameter Golf challenge, targeting a 16MB compressed artifact constraint scored via FineWeb validation bits-per-byte.
- Submitted a result using 13-layer MLA, SmearGate, BigramHash, Muon, and mixed INT5/INT6 quantization: 14.449MB compressed artifact, 1.2838 validation bits per byte before quantization, and 1.3559 after the quantization round trip.

Project Edge | *Raspberry Pi, TFLite, Mosquitto MQTT, Docker, FastAPI*

- Developed a distributed edge-to-cloud IoT pipeline detecting road anomalies in real time on a Raspberry Pi using TFLite inference and streaming telemetry over Mosquitto MQTT.
- Built a Dockerized backend service with FastAPI, LocalStack (S3/DynamoDB), and OwnTracks GPS integration to ingest, deduplicate, and geotag anomalies on an interactive React Leaflet dashboard.

Education

B.Tech, Computer Science & Engineering *Vellore Institute of Technology (VIT)* 2023 - 2027

Chennai, India

CGPA: 8.22 / 10.0 | Relevant coursework: Deep Learning (TensorFlow, CNNs, RNNs, Transformers); Probability and Statistics; Data Analytics.

Awards & honors

1st Place, MongoDB Track; 4th Overall -- DevsHouse25 DevsHouse 2025

Finalist -- Qualcomm Edge AI Hackathon 2025 Qualcomm 2025

Technical skills

Languages: Python, SQL, C, C++, JavaScript, TypeScript, Bash

AI/ML & Data Science: PyTorch, Scikit-Learn, Pandas, NumPy, CUDA, TFLite, HNSW, SimHash, Vector Retrieval, Semantic Caching, Anomaly Detection

Databases & Cloud: PostgreSQL, MySQL, MongoDB, DynamoDB, ChromaDB, AWS (Lambda, S3, Step Functions)

Backend & Systems: FastAPI, REST API design, Docker, Linux, Git, Object-oriented design, Data Analysis, SDLC